SCIENCE FOUNDATIONS

Metrics, Numbers & Unit Analysis

The language every scientist speaks.

PART 1

Numbers Are the Language of Science

What Is Science, Really?

P

It's More Than Math

Science is the study of how nature works. It's full of concepts — motion, energy, forces, waves — and we describe those concepts through math and numbers.

=

Two Halves of the Same Coin

Science is BOTH:

- Qualitative — the concepts and stories
- Quantitative — the math and numbers

You need both to really understand.

Understanding Precision

Which of these is the most precise?

.2	.21	.20788	.20787958	.2078795764
1 sig fig	2 sig figs	3 sig figs	4 sig figs	5 sig figs

⚠ The catch

A number is only as precise as the tool that measured it. Writing .2078795764 when your ruler shows tenths is FALSE PRECISION.

Calculator Trap #1 — Too Many Decimals

Your Calculator

.2078795764

10 digits!

✗ Don't write this down.

What to Do

.208

Rounded to thousandths

✓ Match your data's precision.

Calculator Trap #2 — Scientific Notation on Screen

Calculator Shows

4.587E4

What does that even mean?

✗ Don't write '4.587E4' as your answer.

Write It Two Ways

$$4.587 \times 10^4$$

Scientific notation

OR

45,870

Expanded form

From Atoms to Universes

HUGE

300,000 m

O‘ahu → Big Island

$= 3.0 \times 10^{5}$ m

TINY

.0000003 g

Mass of a mosquito

$= 3.0 \times 10^{-8}$ g

ONE RULE, TWO DIRECTIONS

Scientific Notation

Positive Exponent = BIG

$$3.0 \times 10^{5} = 300{,}000$$

Move the decimal 5 places to the RIGHT.

3.0 → 30 → 300 → 3,000 → 30,000 → 300,000

Or: 'the number of zeros to add.'

Negative Exponent = TINY

$$3.0 \times 10^{-8} = .00000003$$

Move the decimal 8 places to the LEFT.

3.0 → .3 → .03 → ... → .00000003

Or: 'number of decimal shifts.'

PART 2

The Metric System & SI Units

The SI Base Units

M	L	V	T
MASS	LENGTH	VOLUME	TIME
kilogram	**meter**	**liter**	**second**
(kg)	*(m)*	*(L)*	*(s)*
SI unit	SI unit	SI unit	SI unit

Why the Metric System?

ONLY 3

countries

out of ~195

do NOT officially
use the metric system:

United States
Liberia
Myanmar

Why It Matters

Scientists across every continent share data, compare results, and check each other's work. That only works if we all use the same system.

That system is SI (metric). If you want to talk science, you talk metric.

The Metric Ladder — KHDUdcm

King Henry Doesn't Usually Drink Chocolate Milk

K	H	D	U	d	c	m
kilo	hecto	deca	UNIT	deci	centi	milli
×1000	×100	×10	×1	×0.1	×0.01	×0.001
			g / m / L			

← larger *smaller →*

SIX-STEP RECIPE

How to Convert Metric Units

1. **Write the number**

   500 centimeters

2. **Write KHDUdcm**

   K H D U d c m

3. **Mark the starting prefix**

   …c…

4. **Mark the target prefix**

   …U… (m)

5. **Count spaces + direction**

   2 places LEFT

6. **Move the decimal**

   500 → 5.00 m

⚠ *Only works for units that are NOT squared or cubed.*

Example — 500 cm → m

500 cm = ? m

K	H	D	U	d	c	m
			end: U		start: c	

← 2 places LEFT

500. cm → 5.00 m

Practice — Try These

PROBLEM 1

Convert

4,000 m

→

? km

PROBLEM 2

Convert

45.5 L

→

? mL

PART 3

Unit Factor Conversions

Multiplying by 'One' — But Cleverly

1

Anything ÷ Itself = 1

60 minutes = 1 hour

So:

(60 min) ÷ (1 hour) = 1
(1 hour) ÷ (60 min) = 1

Multiplying by 1 doesn't change the value — but it CAN change the units.

→

Example — 2 hours = ? min

2 hours × (60 min / 1 hour)

= 2 × 60 min

= 120 minutes

The 'hours' cancel; the 'minutes' stay.

THREE HABITS THAT MAKE IT FOOLPROOF

Rules for Setting Up Conversions

1

Start with what you have

Include the UNITS. 'Just numbers' is a recipe for errors.

2

Multiply by unit factors

A unit factor is a fraction with equal values in different units (1 cm = 10 mm).

3

Flip factors as needed

Choose the version where the unwanted unit cancels. Cross it out, then move on.

When There's No Direct Equality

How many centimeters are in 1 mile?

Known equalities: 1 inch = 2.54 cm • 1609 m = 1 mile

STEP 1

cm → m
(use KHDUdcm)

1 cm = 0.01 m

STEP 2

m → mile
(unit factor)

1609 m = 1 mile

STEP 3

Multiply chain

1 mile × 1609 m/mile
× 100 cm/m
= 160,900 cm

WHEN TO USE WHICH

Two Methods — At a Glance

M

Metric Conversions

1. Write down the number
2. Write KHDUdcm
3. Mark starting prefix
4. Mark target prefix
5. Count spaces + direction
6. Move the decimal

Best for: single-unit metric jumps (m → km, mg → g)

U

Unit Factor Method

1. Write down what you have (with units!)
2. Find an equality
3. Build unit factors (top or bottom)
4. Multiply so unwanted units cancel
5. Multiply tops, divide bottoms
6. Write final answer with units

Best for: any conversion (metric or not, squared/cubed)

HOMEWORK

Metrics — Problem Sets #1 & #2

Complete the homework in your notebook. Show ALL work — you will be expected to solve the problems on the whiteboard in class. Box or circle your final answer with units.

Homework Problem Sets

classes-list.github.io/middle-school/msp-metrics-problem-sets.html